Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________ 11. ________ 12. ________ 13. ________

Warm-Up: The Discriminant

Example (Warm-Up: The Discriminant):
For $x^2 - 2x - 8 = 0$, the discriminant is $b^2 - 4ac = (-2)^2 - 4(1)(-8) = 4 + 32 = 36$. Since $36 > 0$, there are two real solutions.

Directions: Compute the discriminant $b^2 - 4ac$ for each equation and state the number of real solutions.

1. $x^2 + 4x + 3 = 0$

2. $x^2 - 8x + 16 = 0$

3. $x^2 + 2x + 5 = 0$

4. $2x^2 + 3x - 2 = 0$

Solve with the Formula

Example (Solve with the Formula):
Solve $x^2 - 3x - 10 = 0$. Here $a = 1, b = -3, c = -10$, so $x = \dfrac{3 \pm \sqrt{9 + 40}}{2} = \dfrac{3 \pm 7}{2}$, giving $x = 5$ or $x = -2$.

Directions: Solve each equation using the quadratic formula. Show the discriminant, then simplify completely (leave irrational answers in simplest radical form).

5. $x^2 + 8x + 15 = 0$

6. $x^2 - 9x + 20 = 0$

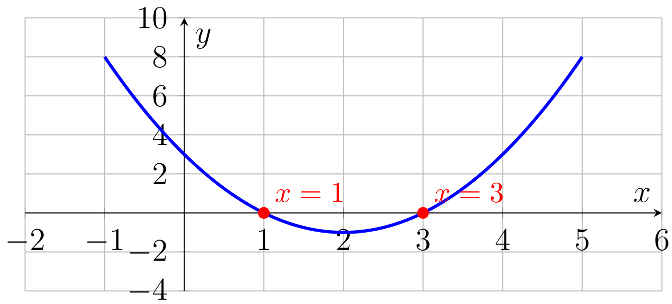
7. $2x^2 + 7x - 4 = 0$

8. $x^2 - 6x + 4 = 0$

9. $3x^2 - 2x - 1 = 0$

Graphical Connection

Example (Graphical Connection):
The parabola $y = x^2 - 2x - 3$ crosses the x-axis at its roots. Solving $x^2 - 2x - 3 = 0$ gives $x = 3$ and $x = -1$, matching the x-intercepts on the graph.



$y = x^2 - 4x + 3$ with roots at $x = 1$ and $x = 3$.

Directions: Use the graph of $y = x^2 - 4x + 3$ to identify the roots, then verify each one using the quadratic formula.

10. Confirm the roots of $x^2 - 4x + 3 = 0$ using the quadratic formula.

11. Compute the discriminant of $x^2 - 4x + 3 = 0$ and explain what its sign tells you about the graph.

Challenge (same sheet):

12. Solve $3x^2 - 8x + 4 = 0$ using the quadratic formula.

13. A ball's height in feet is modeled by $h = -16t^2 + 64t$. Find the times t (in seconds) when the ball is at height $h = 0$.

Answer Key — fully worked

Procedure: Write the equation in standard form $ax^2 + bx + c = 0$.; Identify the values of a, b, and c.; Compute the discriminant $b^2 - 4ac$ to predict the number of real solutions.; Substitute a, b, and c into $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.; Simplify the square root and reduce the fraction to find both solutions.; Check each solution by substituting it back into the original equation..

1. $b^2 - 4ac = 4^2 - 4(1)(3) = 16 - 12 = 4 > 0$ -¿ two real solutions.

2. $b^2 - 4ac = (-8)^2 - 4(1)(16) = 64 - 64 = 0$ -¿ one real (repeated) solution.

3. $b^2 - 4ac = 2^2 - 4(1)(5) = 4 - 20 = -16 < 0$ -¿ no real solutions.

4. $b^2 - 4ac = 3^2 - 4(2)(-2) = 9 + 16 = 25 > 0$ -¿ two real solutions.

5. $a = 1, b = 8, c = 15$; disc $= 64 - 60 = 4$; $x = \dfrac{-8 \pm 2}{2}$ -¿ $x = -3$ or $x = -5$.

6. $a = 1, b = -9, c = 20$; disc $= 81 - 80 = 1$; $x = \dfrac{9 \pm 1}{2}$ -¿ $x = 5$ or $x = 4$.

7. $a = 2, b = 7, c = -4$; disc $= 49 + 32 = 81$; $x = \dfrac{-7 \pm 9}{4}$ -¿ $x = \dfrac{1}{2}$ or $x = -4$.

8. $a = 1, b = -6, c = 4$; disc $= 36 - 16 = 20$; $x = \dfrac{6 \pm \sqrt{20}}{2} = \dfrac{6 \pm 2\sqrt{5}}{2} = 3 \pm \sqrt{5}$.

9. $a = 3, b = -2, c = -1$; disc $= 4 + 12 = 16$; $x = \dfrac{2 \pm 4}{6}$ -¿ $x = 1$ or $x = -\dfrac{1}{3}$.

10. $a = 1, b = -4, c = 3$; disc $= 16 - 12 = 4$; $x = \dfrac{4 \pm 2}{2}$ -¿ $x = 3$ or $x = 1$, matching the x-intercepts shown on the graph.

11. disc $= (-4)^2 - 4(1)(3) = 4 > 0$, which is positive, so the parabola crosses the x-axis at two distinct points (two real roots).

12. $a = 3, b = -8, c = 4$; disc $= 64 - 48 = 16$; $x = \dfrac{8 \pm 4}{6}$ -¿ $x = 2$ or $x = \dfrac{2}{3}$.

13. Set $-16t^2 + 64t = 0$: $a = -16, b = 64, c = 0$; disc $= 64^2 - 4(-16)(0) = 4096$; $t = \dfrac{-64 \pm 64}{-32}$ -¿ $t = 0$ or $t = 4$; since $t = 0$ is the launch, the ball returns to $h = 0$ at $t = 4$ seconds.